Date ___/___/___

ASSIGNMENT - 4

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Section - A

Subject - Database Management System

Ques (1) What is transaction in DBMS?

Ans - A transaction in DBMS is a logical unit of work that consists of one or more database operations such as insert, update or delete, executed as a single unit.

Ques (2) Explain ACID properties.

ACID properties ensure reliable processing of database transactions. They are;

- Atomicity: ensures that a transaction is completed fully or not executed at all.
- Consistency: ensures that a database remains in a valid state before and after the transaction.
- Isolation: ensures that multiple transactions are executed independently without affecting each other.
- Durability: ensures that once a transaction is committed the changes are permanently stored, even in case of system failure.

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Ques (3) What is concurrency control?
Concurrency control in DBMS is the process of managing multiple transactions that are executed simultaneously without causing conflicts or inconsistencies in the database.

Ques (4) What is a deadlock?
A deadlock is a situation in a DBMS where two or more transactions are waiting indefinitely for each other to release resources, and none of them can proceed. Their results in a standstill where the transactions can't complete their execution.

Ques (5) What is serializibility?
It is a property of a database system that ensures the result of executing multiple transactions concurrently is the same as if the transactions were executed one after another (in a serial order).

Ques (6) What is recovery in DBMS?
Recovery in DBMS is the process of restoring the database to a correct and consistent state after a failure, such as system crash, hardware failure or software error. It ensures that no data is lost and that all committed transactions are preserved while incomplete transaction are rolled back.

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Ques (7) What is a checkpoint?

A checkpoint in DBMS is a mechanism that saves the current state of the database at a specific point in time.

It helps reduce recovery time by allowing the system to restart from the last checkpoint instead of processing all transactions from the beginning after a failure.

Design, Analyze and Defend

Ques (8) Explain ACID properties using a banking example.

ACID properties ensures that database transactions are reliable.

Using a banking transfer example [₹500 transferred from Account A to Account B].

- Atomicity: Transaction is all or Nothing.
  - Either ₹500 is deducted from A and added to B, or neither happens.
  - If the system fails after deducting from A, the whole transaction is rolled back.
- Consistency:
  - Total money remains the same
    example - A has ₹2000, B has ₹1000 - After transfer: A has ₹1500, B has ₹1500.
- Isolation:
  - If 2 transactions happen at same time, one transaction won't affect the other's intermediate state.
- Durability: After successful transfer, the updated balances stay saved even if the system crashes.

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Ques (9) Illustrate a deadlock situation with 2 transaction.

A deadlock occurs when two transactions are waiting for each other to release resources and neither compromised.

example -

- Transaction T_1 locks resource A
- Transaction T_2 locks resource B.

Now -

- T_1 tries to lock Resource B (But it is already locked by T_2)
- T_2 tries to lock Resource A (But it is already locked by T_1)

both transactions keep waiting for each other indefinitely, resulting in a deadlock

- Conclusion - Neither T_1, T_2 can continue execution.

Ques (10) Explain serial v/s concurrent execution with example

Serial execution -

In serial execution, transactions are executed one after another, without any overlap. A new transaction starts only after the previous one is completed.

Concurrent execution -

In concurrent execution, multiple transactions are executed at same time, with the operations interleaved to improve preformance and resource utilization.

example - Consider 2 transactions:

- T_1 - Transfer ₹500 from Account A to B.
- T_2 - withdraw ₹200 from Account A.
- In serial, T_1 completes first then T_2 starts.
- In concurrent, operations of T_1 and T_2 may run simultaneously but the final result should be as the same as serial order

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Que (11) A banking system transfers money b/w accounts. Explain how ACID properties apply.

In a banking system, when money is transferred from one account to another, ACID properties ensures safe and reliable transactions.

- Atomicity: The transfer must be fully completed. Money is deducted from one account and added to other. If any step fails, the entire transaction is cancelled.
- Consistency: The total balance remains correct before and after the transaction. The database follows all rules and constraints.
- Isolation: Multiple transactions happening at same time do not interfere with each other. Each transaction behaves as if it is executed alone.
- Durability: Once the transaction is successfully completed, the changes are permanently saved, even if the system crashes.

MARKS (12)